$$\Rightarrow \qquad CP^2 \text{ is } \begin{cases} > r^2 & \text{or,} \\ = r^2 & \text{or} \quad \{CQ = r\}, \\ < r^2. \end{cases}$$

$$\Rightarrow \quad (x_1 + g)^2 + (y_1 + f)^2 \text{ is } \begin{cases} > g^2 + f^2 - c & \text{or,} \\ = g^2 + f^2 - c & \text{or,} \\ < g^2 + f^2 - c. \end{cases}$$

$$\Rightarrow \quad x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c \text{ is } \begin{cases} > 0 & \text{or,} \\ = 0 & \text{or,} \\ < 0. \end{cases}$$

Example 5.5

Examine the position of the point $(2,3)$ with respect to the circle $x^2 + y^2 - 6x - 8y + 12 = 0$.

Solution

Taking (x_1, y_1) as (2, 3), we get

$$x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c = 2^2 + 3^2 - 6 \times 2 - 8 \times 3 + 12,$$
$$= 4 + 9 - 12 - 24 + 12 = -11 < 0.$$

Therefore, the point $(2,3)$ lies inside the circle, by theorem 5.3.

Example 5.6

The line $3x + 4y - 12 = 0$ meets the coordinate axes at A and B. Find the equation of the circle drawn on AB as diameter.

Solution

Writing the line $3x + 4y = 12$, in intercept form yields $\frac{x}{4} + \frac{y}{3} = 1$. Hence the points A and B are $(4,0)$ and $(0,3)$.

Equation of the circle in diameter form is

$$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$$
$$(x - 4)(x - 0) + (y - 0)(y - 3) = 0$$
$$x^2 + y^2 - 4x - 3y = 0.$$

Example 5.7

A line $3x + 4y + 10 = 0$ cuts a chord of length 6 units on a circle with centre of the circle $(2,1)$. Find the equation of the circle in general form.

Solution

$C(2,1)$ is the centre and $3x + 4y + 10 = 0$ cuts a chord AB on the circle. Let M be the midpoint of AB. Then we have

$AM = BM = 3$. Now BMC is a right triangle.

So, we have $CM = \dfrac{|3(2) + 4(1) + 10|}{\sqrt{3^2 + 4^2}} = 4$.

By Pythogoras theorem $BC^2 = BM^2 + MC^2 = 3^2 + 4^2 = 25$.

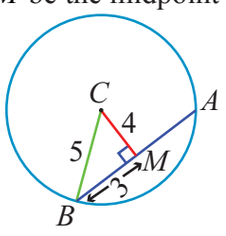


Fig.5.11

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